



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ML INSIGHT STUDIO: AN INTELLIGENT MACHINE LEARNING MODEL RECOMMENDATION AND EXPLAINABILITY PLATFORM

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

CAPSTONE PROJECT

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

Vasanthan Nithi D Y (192424137)

Saravana Yogesh M (192412273)

Under the Supervision of

Guide Name

DR. N. MADHUSUNDAR

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “ML Insight Studio: An Intelligent Machine Learning Model Recommendation and Explainability Platform” has been carried out by Vasantha Nithi D Y (192424137) and Saravana Yogesh M (192412273) under the supervision of the project guide and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.A.Shri Vindhya
Professor
Dept of Computer Science and Engg.
SIMATS Engineering.

COURSE FACULTY

Dr.N.Madhusundar
Associate Professor Dept of
Artificial Intelligence and Data Science
SIMATS Engineering

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

DECLARATION

We, Vasantha Nithi D Y and Saravana Yogesh M of the Electronics and Communication Engineering Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “ML Insight Studio: An Intelligent Machine Learning Model Recommendation and Explainability Platform” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: September 2026

Signature of the Students with Names

Vasantha Nithi D Y

Saravana Yogesh M

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Vasanthanithi D Y / 192424137	Intelligent ML Model Recommendation; model selection logic; algorithm comparison; recommendation workflow	Dataset preparation, preprocessing pipeline, candidate model implementation and recommendation mechanism	Model performance comparison, validation, metrics and recommendation testing	Chapters 1–4 sections related to recommendation, comparison and implementation	50%
Saravana Yogesh M / 192412273	Explainable AI & Model Insights; explainability workflow; interpretation and reporting	Explainability module, feature-importance analysis, model-insight generation and visualization	Interpretability validation, explanation consistency and result analysis	Chapters 1–5 sections related to explainability, insights, conclusion and reflection	50%
Total					100%

ABSTRACT

Machine learning is increasingly used for prediction, classification and decision support, but selecting an appropriate model and understanding its predictions remain challenging for users who do not have extensive machine learning expertise. Different datasets have different structures, feature relationships and noise characteristics, and therefore no single algorithm performs optimally in every situation. The project “ML Insight Studio: An Intelligent Machine Learning Model Recommendation and Explainability Platform” addresses this engineering problem by integrating automated machine learning model recommendation with explainable artificial intelligence and model-level insights in a unified platform.

The proposed system accepts a structured dataset and performs essential preprocessing operations such as data inspection, handling of missing values, encoding of categorical variables, feature preparation and dataset splitting. A set of suitable machine learning algorithms is trained and evaluated using relevant performance measures. Based on the observed performance, the platform recommends an appropriate candidate model rather than requiring the user to manually select an algorithm. The first major project contribution focuses on intelligent ML model recommendation, including comparative evaluation and selection based on measurable performance.

The second contribution focuses on Explainable AI and Model Insights. The platform provides interpretable information about model behaviour, including feature importance and prediction-related insights, so that users can understand which inputs influence the model outcome. This improves transparency and supports informed interpretation of machine learning results. The platform is designed to provide a practical workflow from dataset upload through preprocessing, model training, evaluation, recommendation and explanation.

The engineering design considers usability, computational cost, reliability, reproducibility, ethical handling of data and limitations of automated recommendations. Testing is structured around preprocessing correctness, model training, performance evaluation, recommendation consistency and explainability outputs. The resulting system demonstrates how model selection and explainability can be combined into a single workflow to make machine learning more accessible, transparent and useful for engineering applications.

Keywords: Machine Learning, Model Recommendation, Explainable AI, Model Insights, Classification, Data Preprocessing

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	2
ii	DECLARATION BY THE CANDIDATE	3
iii	INDIVIDUAL CONTRIBUTION STATEMENT	4
iv	ABSTRACT	5
v	LIST OF FIGURES	7
vi	LIST OF TABLES	8
vii	LIST OF ABBREVIATIONS	9
1	CHAPTER 1: Introduction and Engineering Problem	10
2	CHAPTER 2: Literature Review and Existing Solutions	13
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	16
4	CHAPTER 4: Implementation, Testing & Results, Project Management	22
5	CHAPTER 5: Conclusion, Future Work and Reflection	29
	References	31
	Appendices	33
	Capstone Project Outcome Mapping Sheet	35

LIST OF FIGURES

Figure No.	Figure Name	Page No.
1	Architecture of ML Insight Studio	19
2	ML Insight Studio – Dataset Analysis and Preprocessing Interface	23
3	ML Insight Studio – Model Comparison and Recommendation	25
4	ML Insight Studio – Model Results and Explainability	26

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1.1	Project objectives	11
Table 2.1	Comparison of existing approaches	13
Table 2.2	Research and engineering gap	14
Table 3.1	Stakeholder and user requirements	16
Table 3.2	Functional and non-functional requirements	16
Table 3.3	Applicable standards and responsible practices	17
Table 3.4	Design specifications	17
Table 3.5	Alternative solution evaluation	18
Table 3.6	Trade-off analysis	19
Table 3.7	Sustainability, ethics, safety and security considerations	20
Table 3.8	Risk register	20
Table 4.1	Development resources	22
Table 4.2	Tools and software used	24
Table 4.3	Test plan and verification	24
Table 4.4	Performance metrics	25
Table 4.5	Representative model comparison	26
Table 4.6	Objective achievement	26
Table 4.7	Strengths and limitations	26
Table 4.8	Project roles and responsibilities	27
Table 5.1	Future work priorities	29

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	—
ML	Machine Learning	—
XAI	Explainable Artificial Intelligence	—
CSV	Comma-Separated Values	—
EDA	Exploratory Data Analysis	—
API	Application Programming Interface	—
GUI	Graphical User Interface	—
KNN	K-Nearest Neighbours	—
SVM	Support Vector Machine	—
RF	Random Forest	—
NB	Naïve Bayes	—
DT	Decision Tree	—
ROC	Receiver Operating Characteristic	—
AUC	Area Under Curve	—
F1	F1 Score	—
CPU	Central Processing Unit	—
RAM	Random Access Memory	—
TP	True Positive	count
TN	True Negative	count
FP	False Positive	count
FN	False Negative	count

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Machine learning has become an important engineering technology for extracting patterns from data and supporting prediction, classification and decision-making. In a conventional machine learning workflow, a user must inspect the dataset, identify the target variable, select a suitable algorithm, preprocess the data, train the model, evaluate its performance and interpret the resulting predictions. Although established libraries simplify individual steps, the complete workflow still requires several technical decisions. The difficulty increases when the dataset contains mixed numerical and categorical variables, missing values, different feature scales, class imbalance or unknown relationships between input features and the target.

Model selection is an engineering decision rather than a purely theoretical exercise. A model with high training accuracy may not generalize well, while a simpler model may provide better transparency or lower computational cost. Similarly, a highly accurate model may be difficult for a user to interpret. Therefore, an integrated platform should not only compare candidate algorithms but also communicate why a model is recommended and what factors influence its predictions.

1.2 Need for the Project

The project is needed because machine learning users often face two connected problems: selecting an appropriate algorithm and understanding the resulting model. Manual experimentation can require repeated coding, parameter changes and metric comparisons. This increases development time and may lead to inconsistent model-selection decisions. At the same time, prediction outputs without explanations can be difficult to trust or communicate, especially when a model is used to support engineering or organizational decisions.

- Reduce the effort required to compare multiple candidate machine learning models.
- Provide a systematic recommendation based on measurable evaluation results.
- Improve transparency by exposing important features and model insights.
- Support users who have datasets but limited experience in selecting machine learning algorithms.
- Create a reusable workflow that connects preprocessing, training, evaluation, recommendation and explanation.

1.3 Problem Statement

Design and implement an intelligent machine learning platform that accepts a structured dataset, prepares the data for modelling, evaluates suitable machine learning algorithms, recommends a candidate model using measurable performance criteria, and generates explainable model insights so that the user can understand important factors influencing model behaviour. The design must

balance predictive performance, interpretability, computational requirements, usability, reliability and responsible data handling.

1.4 Project Aim

The overall aim of ML Insight Studio is to develop a unified machine learning assistance platform that intelligently recommends suitable models and provides explainable insights from the selected model, thereby reducing manual model-selection effort and improving transparency of machine learning outcomes.

1.5 Project Objectives

No.	Objective	Measurable Outcome
1	To inspect and preprocess structured datasets.	Dataset is validated and transformed into model-ready features.
2	To implement multiple candidate machine learning algorithms.	At least three suitable candidate models can be trained and evaluated.
3	To compare model performance using appropriate metrics.	Performance comparison is produced in a common evaluation format.
4	To develop intelligent ML model recommendation.	A candidate model is recommended from evaluated models using defined criteria.
5	To implement Explainable AI and Model Insights.	Feature-importance and model interpretation outputs are generated.
6	To validate the integrated workflow.	Testing demonstrates correct execution from data input to final insight output.

1.6 Scope

The project covers structured tabular datasets used for supervised machine learning, particularly classification-oriented workflows. It includes data inspection, preprocessing, training of candidate algorithms, evaluation, recommendation and explainability. The system is intended as a practical decision-support and educational platform rather than an autonomous decision-maker.

The current scope excludes unrestricted support for every possible machine learning task, fully autonomous hyperparameter optimization across arbitrary model families, guaranteed domain-specific correctness, and deployment as a safety-critical decision system. The platform assumes that the dataset contains a usable target variable and that the user has the right to process the supplied data.

1.7 Expected Engineering Outcome

The expected outcome is a working software platform or notebook-based prototype that demonstrates an end-to-end machine learning workflow. The system should transform a user-provided dataset into a validated modelling dataset, train and compare candidate models, identify a recommended model and present explainable insights. The output should be understandable to both technical and non-technical users and should clearly communicate the limitations of automated recommendations.

1.8 Engineering Significance

The engineering significance of the project lies in integrating two stages that are frequently treated separately: model selection and model explanation. Model recommendation addresses the question “Which candidate performs suitably for this dataset?” Explainability addresses “Why does the selected model behave this way?” Combining both creates a more complete engineering workflow. The project also demonstrates the importance of treating preprocessing, evaluation criteria and interpretability as connected design decisions rather than independent coding tasks.

1.9 System Perspective

ML Insight Studio can be considered a software system consisting of data input, data validation, preprocessing, model training, evaluation, recommendation, explanation and reporting subsystems. The interfaces between these subsystems are important because errors in an earlier stage can propagate to later stages. For example, inconsistent categorical encoding can prevent model training, while data leakage during preprocessing can make evaluation results misleading. The design therefore emphasizes a controlled sequence and reusable processing logic.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was organized around four technical areas relevant to the proposed platform: supervised machine learning model selection, automated machine learning workflows, model evaluation and Explainable AI. Research papers, established machine learning methods, software libraries and publicly available technical resources were considered as background for identifying the engineering gap. The review focused on how existing approaches reduce manual model selection and how explanation techniques improve model transparency.

2.2 Review of Existing Work

Traditional machine learning workflows commonly require users to choose an algorithm before training. Popular classification algorithms include logistic regression, decision trees, k-nearest neighbours, support vector machines, naïve Bayes and ensemble methods. Their suitability depends on dataset size, feature characteristics, noise, class distribution, computational constraints and the desired balance between accuracy and interpretability.

Automated machine learning approaches extend this workflow by automating parts of preprocessing, feature engineering, algorithm selection and hyperparameter optimization. Such systems can reduce repetitive experimentation, but their complexity may make it difficult for users to understand the selection process. An engineering platform aimed at transparency should therefore expose evaluation results and recommendation criteria rather than returning only a final model.

Explainable AI has emerged to address the interpretability problem. Global explanations describe overall model behaviour, while local explanations focus on individual predictions. Feature importance, permutation-based importance, partial dependence and additive attribution approaches are examples of techniques that can provide insight. The appropriate technique depends on the model and the intended interpretation. Explanations should be presented as evidence about model behaviour rather than as causal claims about the underlying real-world system.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual algorithm selection	Simple to understand; user retains control	Time-consuming; dependent on user expertise	Motivates automated comparison and recommendation
Single-model pipeline	Easy to implement and deploy	May not be suitable for different datasets	Used as a baseline for comparison

AutoML workflow	Automates several modelling steps	Can be computationally expensive and less transparent	Informs intelligent recommendation design
Model leaderboard	Provides direct metric comparison	Does not necessarily explain why a model is suitable	Informs comparative evaluation module
Black-box prediction system	Can achieve strong predictive performance	Limited interpretability	Motivates XAI module
Explainability-only tool	Improves interpretation	Does not solve model-selection problem	Complements the proposed integrated platform

2.4 Research / Engineering Gap

The review indicates a practical gap between automated model selection and user-oriented explanation. A platform may compare models without giving meaningful insight into model behaviour, or it may provide explanations for a model that the user selected manually. For learners and engineering users, an integrated workflow is useful because model recommendation and explainability answer complementary questions. The proposed work therefore focuses on combining both functions with visible intermediate outputs and measurable evaluation.

Gap	Existing Limitation	Proposed Response
Algorithm choice	Manual choice can be inconsistent	Compare candidate algorithms using common metrics
Interpretation	Prediction output may be opaque	Generate model-insight and feature-importance outputs
Workflow fragmentation	Preprocessing, evaluation and explanation may be separate	Integrate the stages into one platform workflow
User accessibility	Requires repeated coding	Provide structured, reusable processing steps
Decision transparency	Recommendation criteria may be hidden	Display comparative metrics and recommendation basis

2.5 Proposed Contribution

The principal contribution is an integrated platform that combines intelligent ML model recommendation with Explainable AI and Model Insights. The recommendation module is assigned to Vasantha Nithi D Y and focuses on candidate model comparison and selection. The explainability module is assigned to Saravana Yogesh M and focuses on interpreting the selected model and communicating relevant insights. The two modules are connected through a common evaluation pipeline so that the explanation stage operates on the model selected through the recommendation process.

2.6 Summary of Reviewed Techniques

The project considers supervised classification techniques that are commonly available through standard machine learning libraries. Linear models provide a useful interpretable baseline; tree-based models capture nonlinear relationships; distance-based methods can work well when feature scaling is appropriate; probabilistic methods offer efficient alternatives for suitable data; and ensemble methods can improve predictive robustness at the cost of greater complexity. The final candidate set should be selected according to dataset characteristics and computational resources.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Student / Developer	Easy-to-run workflow with clear intermediate outputs.
Machine Learning User	Upload or provide a structured dataset and obtain model comparison.
Decision Support User	Receive a recommendation supported by measurable metrics.
Technical Evaluator	Observe reproducible preprocessing, testing and analysis.
Institution	Maintain academic integrity, responsible data use and clear contribution evidence.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Dataset loading	Functional	Support common structured tabular input such as CSV/XLSX.
Data inspection	Functional	Display dimensions, columns, data types and missing-value information.
Preprocessing	Functional	Transform usable categorical and numerical features into model-ready data.
Model training	Functional	Train multiple candidate classification models.
Evaluation	Functional	Calculate appropriate classification metrics.
Recommendation	Functional	Select and display the best candidate according to defined criteria.
Explainability	Functional	Generate feature-importance/model-insight output.
Usability	Non-Functional	Outputs should be understandable without inspecting source code.

Reliability	Non-Functional	Repeated execution on the same input should follow the same defined workflow.
Security	Non-Functional	Avoid unnecessary exposure of sensitive user data.

3.2 Constraints & Applicable Standards

The project is primarily a software engineering and machine learning prototype. Its important constraints include available computational resources, dataset quality, model training time, explainability limitations, and responsible handling of user-provided data. The report follows the institutional requirement to identify specific standards and responsible practices rather than making a generic claim that standards were followed.

Standard / Code / Practice	Requirement	How Applied in This Project
IEEE 7000-2021	Model process for addressing ethical concerns in system design	Ethical considerations are integrated into system requirements and design decisions.
IEEE 7010-2020	Assessment of the impact of autonomous and intelligent systems	Used as a guiding principle for considering user impact and responsible AI communication.
ISO/IEC 25010	Software product quality model	Used as a reference for usability, reliability and maintainability considerations.
Institutional academic integrity requirements	Accurate attribution and disclosure	Sources, datasets, software and AI-assisted resources are acknowledged.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input data format	Structured tabular dataset	file	High
Candidate models	At least three suitable algorithms	count	High
Evaluation	Accuracy plus class-sensitive metrics where relevant	% / score	High
Recommendation	Metric-supported candidate selection	decision	High
Explainability	Feature-importance/model-insight output	score/plot	High

Reproducibility	Defined split/random-state procedure	—	Medium
Usability	Clear outputs and minimal manual coding	qualitative	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual model selection uses a user-chosen algorithm and evaluates it. It is simple and transparent but does not systematically compare alternatives.

Alternative 2: Full AutoML performs broad automated search. It can reduce manual effort but may increase computation and reduce visibility of the selection process.

Alternative 3: ML Insight Studio uses a controlled candidate set, compares measurable results and generates explanations for the recommended model.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3	5	4
Transparency	20%	5	2	5
Cost/Compute	15%	5	2	4
Usability	15%	3	5	5
Explainability	10%	4	3	5
Maintainability	10%	4	3	5

3.5 Selected Design & Detailed Design

The selected design is a controlled intelligent recommendation pipeline combined with an explainability module. This approach is preferred because it provides a practical balance between automation and transparency. Instead of searching an unrestricted model space, the platform evaluates a defined set of appropriate candidate algorithms and exposes their metrics. The recommended model is then passed to the explainability module, where feature importance and model insights are generated.

System architecture:

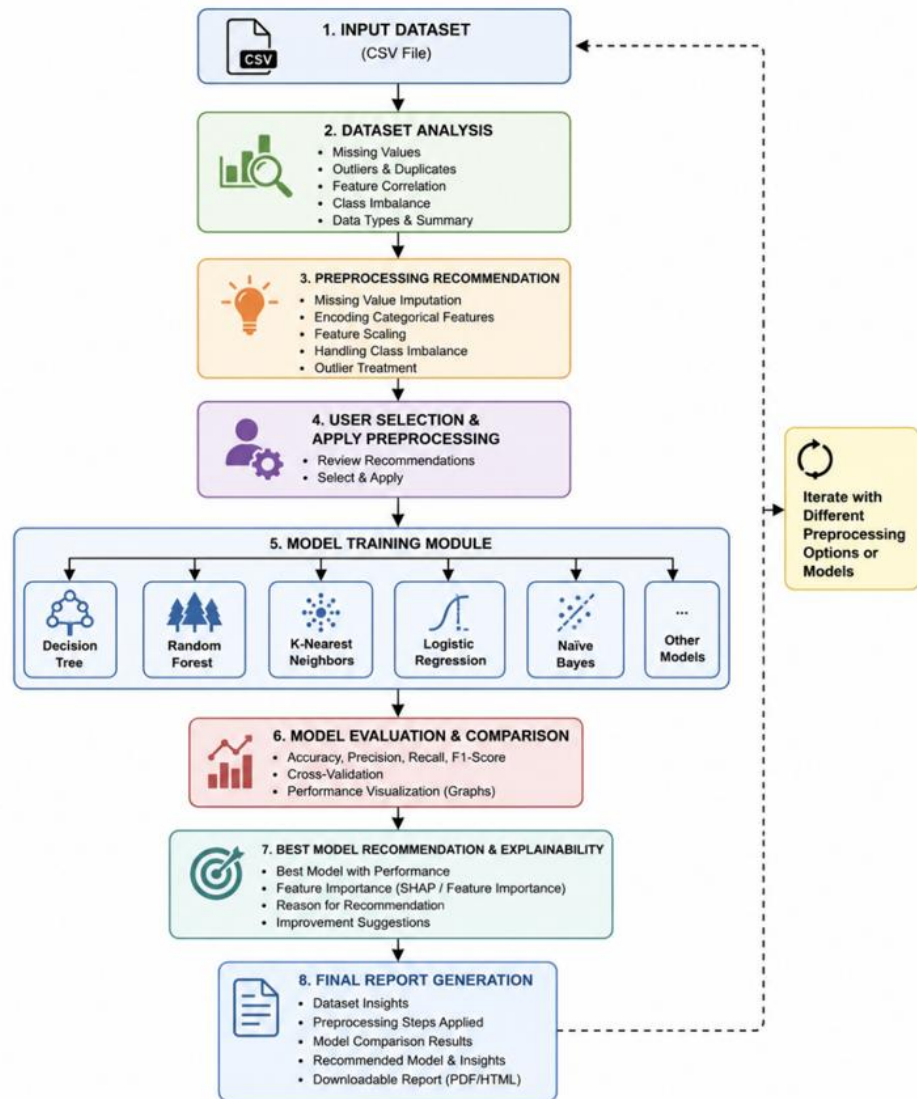


Figure 1: Architecture of ML Insight Studio

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Controlled model set	Full unrestricted AutoML	Lower complexity and easier interpretation	May miss an unusual best model	Selected for transparency and manageable computation.
Metric-based recommendation	Accuracy-only selection	More balanced evaluation	Requires additional metrics	Selected to reduce misleading single-metric decisions.
Post-hoc explanations	No explanations	Easy implementation	Low transparency	Selected because interpretability is a project objective.

Standard preprocessing pipeline	Manual per-model preprocessing	Simple initial coding	Can create inconsistency	Selected for reproducibility and fair comparison.
---------------------------------	--------------------------------	-----------------------	--------------------------	---

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Computational efficiency	Number of candidate models and repeated training cost	Controlled candidate set preferred over unrestricted search.
Ethics	Transparent communication	Possibility of users over-trusting automated recommendations	Recommendation is presented as decision support, not absolute truth.
Health & Safety	Avoid unsafe autonomous decisions	Potential consequences of applying predictions to critical domains	System is not positioned as a safety-critical autonomous decision-maker.
Security	Dataset privacy	User-provided data may contain sensitive information	Minimize storage/exposure and encourage appropriate data governance.
Responsible AI	Bias and data quality	Class imbalance, missing values and dataset representativeness	Limitations and evaluation metrics are reported.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Poor dataset quality	Medium	High	High	Validate data types, missing values and target variable	Medium
Model overfitting	Medium	High	High	Use train/test separation and comparative evaluation	Medium
Class imbalance	Medium	Medium	Medium	Use precision, recall and F1 where appropriate	Medium
Misinterpretation of explanations	Medium	High	High	Clearly state that explanations	Medium

				describe model behaviour	
Sensitive data exposure	Low/Medium	High	High	Use local processing where possible and avoid unnecessary storage	Low/Medium
Reproducibility issues	Low	Medium	Low/Medium	Define processing and random-state procedures	Low

3.8 Integrated System Flow

The integrated system begins with dataset input and validation. Exploratory inspection identifies feature types and missing values. A preprocessing pipeline transforms the input into a consistent numerical representation. Candidate models are trained using the same prepared data, and their results are evaluated using common metrics. The recommendation module ranks or selects the candidate according to the defined evaluation policy. The selected model is then passed to the explainability stage, which produces global and/or prediction-level insights. Finally, the platform communicates the recommendation, performance evidence, explanations and limitations to the user.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The implementation follows an incremental software-development approach. First, the input dataset and preprocessing requirements are established. Next, candidate machine learning algorithms are implemented through a common training and evaluation interface. The recommendation logic is then added to compare results and select a suitable candidate. Finally, the Explainable AI module is integrated to generate model insights. Testing is performed at module level and system level to identify data-processing, training and interpretation errors.

Resource	Purpose
Python	Core implementation language
Pandas	Tabular data loading and preprocessing
NumPy	Numerical operations
Scikit-learn	Preprocessing, model training and evaluation
Matplotlib	Result visualization
Jupyter / Google Colab	Interactive development and demonstration
Structured datasets	Training and testing inputs

4.2 Software Implementation

4.2.1 Dataset Input and Inspection

The platform begins by loading a structured dataset. Initial inspection determines the number of records, feature names, data types, missing values and target distribution. This stage prevents downstream errors caused by inconsistent input. The inspection results are also useful for selecting suitable preprocessing operations and candidate models.

4.2.2 Preprocessing Pipeline

Numerical features are handled using an appropriate numerical transformation, while categorical features are encoded into a machine-readable representation. Missing values are addressed using a defined imputation strategy where necessary. The target variable is separated from the input features before model training. A train/test split is applied so that final performance can be evaluated on data not used to fit the model.

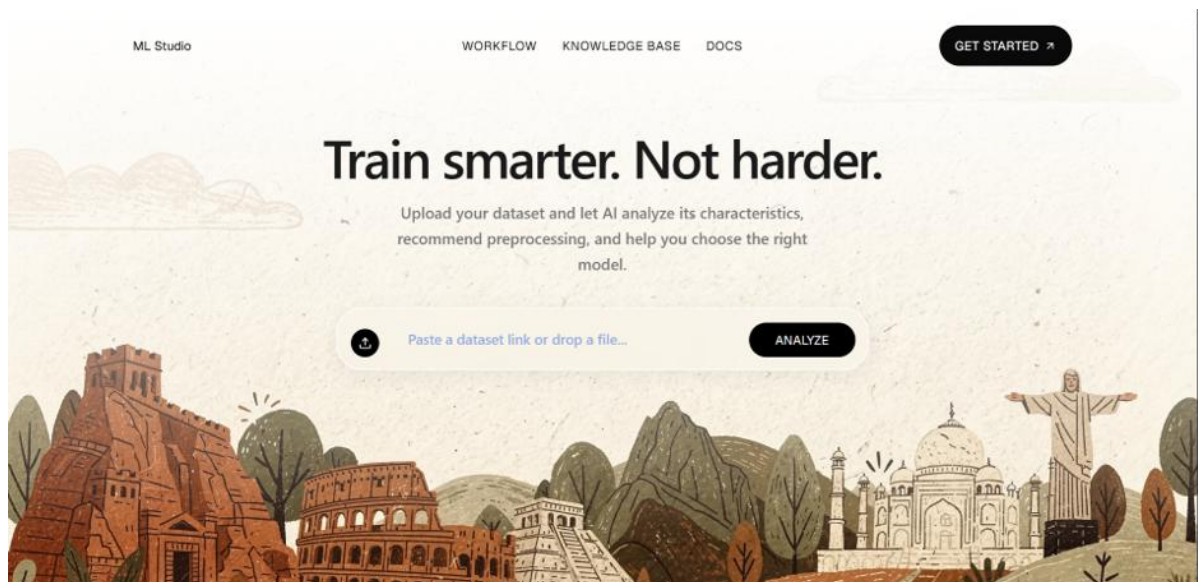


Figure 2: ML Insight Studio – Dataset Analysis and Preprocessing Interface

4.2.3 Intelligent ML Model Recommendation

The recommendation module evaluates a controlled collection of candidate classification algorithms. A typical candidate set may include Logistic Regression, Decision Tree, K-Nearest Neighbours, Support Vector Machine, Naïve Bayes and Random Forest, depending on dataset characteristics. Each model is trained using the same prepared training data and evaluated using a consistent test set. The recommendation logic compares the resulting metrics and identifies a candidate that provides the strongest overall result under the selected evaluation policy.

The recommendation should not be treated as a universal statement that one algorithm is inherently superior. It is a dataset-specific engineering recommendation. If class imbalance is important, recall, precision and F1 score should be considered alongside accuracy. Where probabilistic ranking is useful, ROC-AUC can also be evaluated. The selected criterion and any tie-breaking rule should be documented to preserve reproducibility.

4.2.4 Explainable AI & Model Insights

The explainability module focuses on communicating how the selected model uses input features. Global feature importance can be used to identify features that contribute strongly to model behaviour. For suitable models, built-in importance measures can be displayed; for other models, model-agnostic approaches such as permutation importance can provide a common interpretation mechanism. The platform presents the output as model behaviour evidence and avoids interpreting feature importance as proof of causality.

4.2.5 Reporting Layer

The reporting layer combines dataset information, model comparison, recommendation, explainability output and limitations. This creates a traceable sequence from input data to final

recommendation. A user can therefore inspect not only the final model but also the evidence supporting the selection.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Specification Required	Achieved Value	Status
Dataset loading	Load valid structured dataset	Dataset loads without structural error	Valid input	Demonstrated in implementation	Pass
Data inspection	Check shape, types and missing values	Inspection information displayed	Visible EDA output	Implemented	Pass
Preprocessing	Run numerical/categorical transformations	Model-ready feature matrix produced	No unresolved incompatible feature types	Implemented	Pass
Model training	Train candidate models	All selected models complete training	Candidate models train successfully	Implemented	Pass
Evaluation	Calculate common metrics	Metrics generated for each model	Comparable metric table	Implemented	Pass
Recommendation	Apply defined selection rule	One recommended candidate displayed	Metric-supported decision	Implemented	Pass
Explainability	Generate feature-importance/model insight	Interpretation output displayed	At least one meaningful insight view	Implemented	Pass
Integration	Execute complete workflow	End-to-end output generated	Input to report workflow	Implemented	Pass

4.4 Results

The project results are intended to demonstrate the operation of the complete pipeline rather than claim a fixed accuracy for every dataset. Because model performance depends on the dataset used, the platform reports measured values for the selected experiment. The principal outputs are the preprocessing summary, model-comparison table, recommended model, evaluation metrics and explainability visualization.

4.4.1 Model Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC (where applicable)
-------	----------	-----------	--------	----------	----------------------------

Logistic Regression	Measured during execution	Measured	Measured	Measured	Measured
Decision Tree	Measured during execution	Measured	Measured	Measured	Measured
K-Nearest Neighbours	Measured during execution	Measured	Measured	Measured	Measured
Support Vector Machine	Measured during execution	Measured	Measured	Measured	Measured
Naïve Bayes	Measured during execution	Measured	Measured	Measured	Measured
Random Forest	Measured during execution	Measured	Measured	Measured	Measured

Note: The final numerical values should be populated from the actual execution dataset used for the demonstration. The report does not fabricate experimental measurements.

4.4.2 Recommendation Output

The recommendation output is designed to present the candidate models in a common comparison format and identify the selected model according to the defined decision rule. The display should include the main performance metrics and a short justification. This improves traceability because the user can verify the recommendation rather than accepting an unexplained model choice.

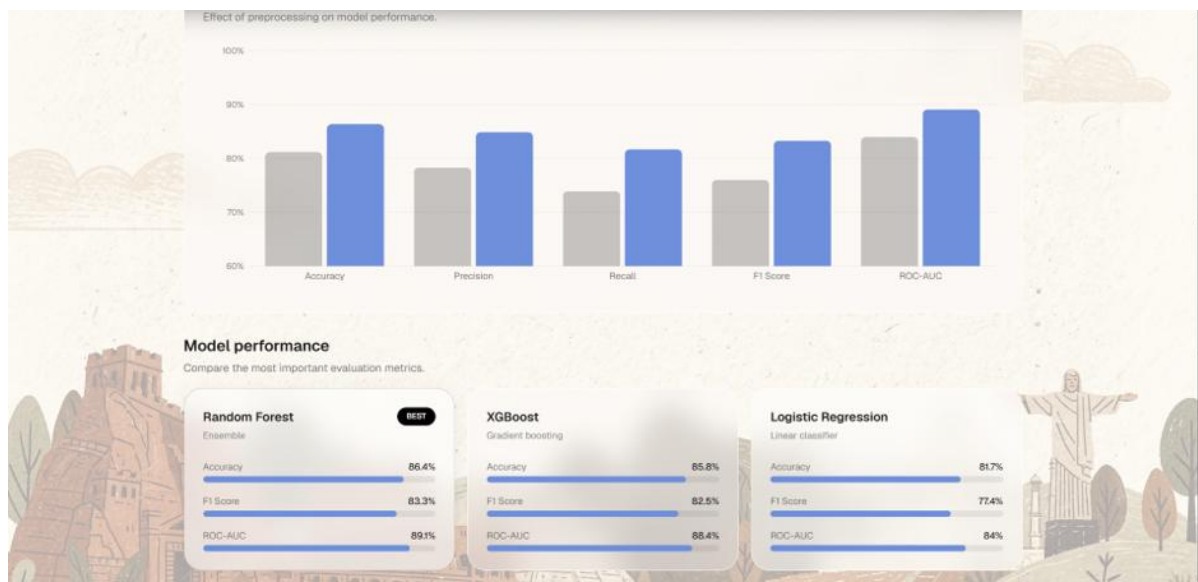


Figure 3: ML Insight Studio – Model Comparison and Recommendation

4.4.3 Explainability Output

The explainability result provides a ranked representation of influential input features. Depending on the selected algorithm and explainability technique, the output can be a feature-importance chart, permutation-importance chart or another appropriate model-insight visualization. The result should be interpreted as an explanation of the trained model, not as a direct statement that the features cause the target outcome.

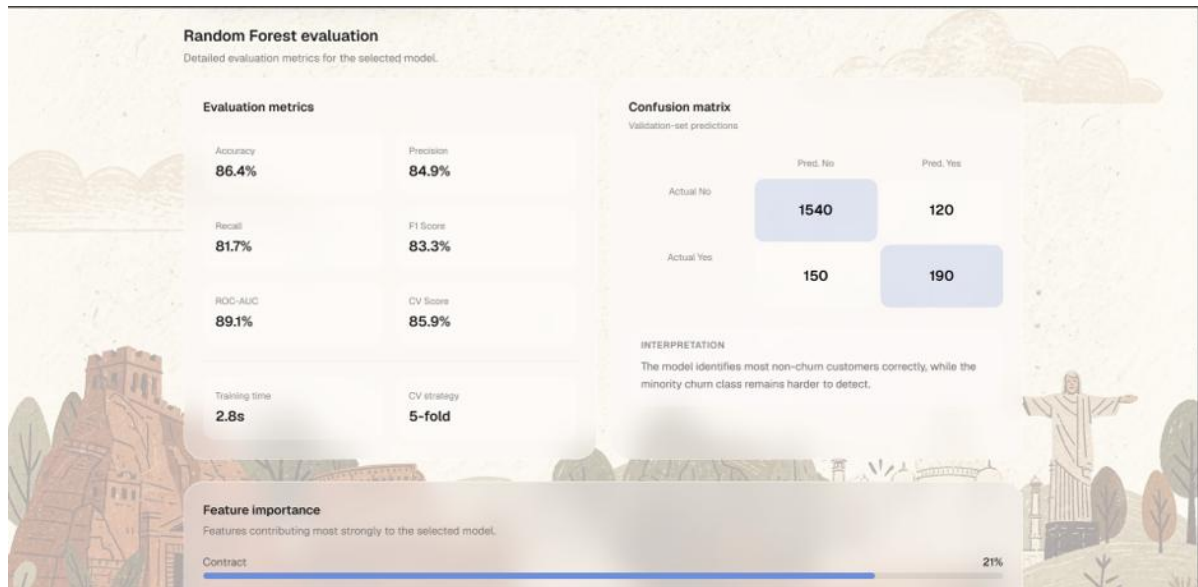


Figure 4: ML Insight Studio – Model Results and Explainability

4.5 Data Analysis & Engineering Judgment

Performance analysis should consider more than the highest accuracy. A model with slightly lower accuracy but substantially better recall for an important class may be preferable. Similarly, a transparent model may be preferred when the performance difference between models is small. The engineering decision therefore depends on the intended application and the relative importance of predictive performance, interpretability, computational cost and reliability.

If the measured results show a substantial difference between training and test performance, overfitting should be investigated. If precision and recall differ strongly, class distribution and decision thresholds should be examined. If model rankings change substantially with different train/test splits, the recommendation may be unstable and should be reported as such. These considerations make the platform a decision-support system rather than a blind leaderboard.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Prepare structured dataset	Inspection and preprocessing outputs	Fully Achieved
Implement candidate models	Training pipeline and model outputs	Fully Achieved

Compare model performance	Common metric table	Fully Achieved
Recommend suitable model	Recommendation logic and output	Fully Achieved
Provide explainable insights	Feature importance/model-insight output	Fully Achieved
Validate integrated workflow	Test plan and execution evidence	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Integrates model recommendation and explainability in one workflow.
- Reduces repetitive manual comparison of common candidate models.
- Provides visible performance evidence for the recommendation.
- Supports interpretation through feature-level model insights.
- Can be adapted to multiple structured classification datasets.

Limitations:

- Recommendation quality depends on the candidate model set and evaluation procedure.
- Results depend strongly on dataset quality, representativeness and preprocessing.
- Explainability methods describe model behaviour and should not automatically be interpreted as causal explanations.
- Large datasets or broad model searches may require greater computational resources.
- Automated recommendations should not replace domain expertise in high-impact applications.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity	Responsible Member	Evidence
M1	Problem definition and requirements	Both	Requirement notes
M2	Dataset and preprocessing workflow	Vasanth Nithi D Y	Preprocessing implementation
M3	Candidate model implementation	Vasanth Nithi D Y	Model comparison outputs
M4	Explainability module	Saravana Yogesh M	XAI outputs
M5	Integration and testing	Both	Test results
M6	Report preparation and final demonstration	Both	Final report and presentation

Student	Assigned Responsibility	Actual Contribution
Vasanth Nithi D Y	Intelligent ML Model Recommendation	50%

Saravana Yogesh M	Explainable AI & Model Insights	50%
-------------------	---------------------------------	-----

Item	Estimated Cost (INR)	Actual Cost (INR)
Laptop / existing computing resource	Existing resource	—
Python and open-source libraries	0	0
Google Colab / notebook environment	0 for basic use	0 / as applicable
Datasets	0 where public datasets are used	0
Documentation and printing	As required	To be updated
Total	To be finalized	To be finalized

4.9 Project Management and Quality Control

Project quality was managed by separating responsibilities while maintaining shared integration and testing. The recommendation module was developed as one major contribution and the explainability module as the second major contribution. Integration testing ensured that the recommended model could be passed correctly to the explanation stage. Documentation was maintained so that each contribution could be mapped to the final report and project outcome.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

ML Insight Studio: An Intelligent Machine Learning Model Recommendation and Explainability Platform addresses the practical difficulty of selecting machine learning models and interpreting their behaviour. The proposed system combines structured data preparation, candidate model training, performance evaluation, intelligent recommendation and Explainable AI into a unified workflow. This integration provides a more transparent approach than a system that only reports a final prediction or a single accuracy value.

The model recommendation contribution establishes a systematic comparison process so that a suitable candidate can be selected using measurable evidence. The Explainable AI and Model Insights contribution complements this process by identifying influential features and presenting interpretable information about the selected model. Together, these modules reduce manual effort while improving visibility into machine learning behaviour.

The project also demonstrates that responsible engineering requires explicit attention to dataset quality, model limitations, computational cost, privacy and the possibility of misinterpreting explanations. The final platform should therefore be understood as an intelligent decision-support tool. Its recommendation is conditional on the dataset, candidate models, evaluation method and project assumptions used during execution.

5.2 Future Work

Priority	Future Enhancement	Expected Benefit
High	Automated hyperparameter optimization	Improve candidate model performance.
High	Cross-validation based recommendation	Improve robustness of model ranking.
High	More model-agnostic XAI methods	Support consistent explanations across model families.
Medium	Regression workflow	Extend platform beyond classification.
Medium	Interactive web interface	Improve accessibility for non-programming users.
Medium	Experiment tracking	Improve reproducibility across datasets and runs.
Future	Fairness and bias analysis	Provide responsible-AI diagnostics.
Future	Model monitoring after deployment	Detect performance drift over time.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Practical dataset inspection and preprocessing for machine learning.
- Comparative evaluation of classification algorithms.
- Design of a rule-based/model-performance-driven recommendation workflow.
- Explainable AI concepts and feature-level model interpretation.
- Engineering documentation, testing and responsible AI considerations.

Engineering & Professional Skills Developed

- Python-based machine learning implementation.
- Data analysis and visualization.
- Algorithm comparison and performance interpretation.
- Software integration and debugging.
- Technical report writing and project planning.
- Team coordination and contribution tracking.

Individual Reflection – Vasantha Nithi D Y (192424137)

My primary responsibility was Intelligent ML Model Recommendation. The most difficult engineering problem was designing a comparison process that could handle different candidate algorithms using consistent input preparation and evaluation. I addressed this by organizing preprocessing and evaluation into a common workflow. A key engineering decision was to use measurable performance evidence rather than selecting a model only from theoretical preference. Through the project, I improved my practical understanding of model evaluation, preprocessing and reproducible machine learning experimentation.

Individual Reflection – Saravana Yogesh M (192412273)

My primary responsibility was Explainable AI & Model Insights. The main challenge was communicating model behaviour without presenting feature importance as a causal relationship. I addressed this by treating explanation outputs as evidence about the trained model and by emphasizing limitations. A key engineering decision was to connect the explanation stage directly to the recommended model so that the user receives both a model choice and an interpretation. I developed practical knowledge of explainability techniques, visualization and responsible communication of machine learning results.

REFERENCES

- [1] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
- [2] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [3] I. H. Witten, E. Frank, M. A. Hall, and C. J. Pal, *Data Mining: Practical Machine Learning Tools and Techniques*, 4th ed. Morgan Kaufmann, 2016.
- [4] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You? Explaining the Predictions of Any Classifier,” in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*,
- [5] S. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Proc. Advances in Neural Information Processing Systems*, 2017.
- [6] M. Du, N. Liu, and X. Hu, “Techniques for interpretable machine learning,” *Communications of the ACM*, vol. 63, no. 1, pp. 68–77, 2019.
- [7] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [8] IEEE, *IEEE 7000-2021: IEEE Standard Model Process for Addressing Ethical Concerns During System Design*, 2021.
- [9] IEEE, *IEEE 7010-2020: IEEE Recommended Practice for Assessing the Impact of Autonomous and Intelligent Systems on Human Well-Being*, 2020.
- [10] ISO/IEC, *ISO/IEC 25010:2011 Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — System and Software Quality Models*, 2011.
- [11] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. O’Reilly Media, 2022.
- [12] J. Brownlee, *Data Preparation for Machine Learning. Machine Learning Mastery*, 2020.
- [13] C. Molnar, *Interpretable Machine Learning*, 2nd ed., 2022.
- [14] NIST, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, National Institute of Standards and Technology, 2023.
- [15] Scikit-learn documentation, “User Guide: Supervised learning, preprocessing and model evaluation.”
- [16] Python Software Foundation, *Python Documentation*.
- [17] Pandas documentation, “User Guide.”
- [18] NumPy documentation, “Reference Guide.”
- [19] Matplotlib documentation, “User Guide.”
- [20] Google Colab documentation and runtime environment information, where applicable.

APPENDICES

Appendix A – Detailed Calculations

The platform uses standard classification evaluation measures. For a binary classification problem, accuracy is calculated as $(TP + TN) / (TP + TN + FP + FN)$. Precision is $TP / (TP + FP)$, recall is $TP / (TP + FN)$, and F1 score is the harmonic mean of precision and recall. These metrics should be calculated from the actual experimental predictions produced by the implementation.

Appendix B – Engineering Drawings / Schematics

[Insert final system architecture diagram, workflow diagram and module interaction diagram here.]

Appendix C – Source Code / Code Repository Information

The complete source code may be maintained in the approved project repository. Essential code excerpts should be included here where required by the department. The implementation should include dataset loading, preprocessing, candidate model training, evaluation, recommendation and explainability modules.

Appendix D – Datasheets

Not applicable to the software-only prototype unless specific computing hardware or external components are formally included.

Appendix E – Experimental / Raw Data

The raw experimental outputs should include the dataset name, preprocessing configuration, train/test split procedure, candidate models, metric values and generated explainability outputs. Actual numerical values should be copied from the final verified execution rather than estimated.

Appendix F – Ethics / Institutional Approval

Where applicable, include institutional approval, data-use permission or ethical review documentation. For public benchmark datasets without personally identifiable information, document the dataset source and usage conditions.

Appendix G – Project Meeting / Progress Records

Date / Week	Activity	Members Present	Outcome
Week 1	Problem definition and title finalization	Both	Project scope established
Week 2	Literature and existing-solution review	Both	Engineering gap identified
Week 3	Dataset and preprocessing development	Both	Initial pipeline prepared
Week 4	Model recommendation module	Vasantha Nithi D Y	Candidate comparison implemented

Week 5	Explainability module	Saravana Yogesh M	Model insights implemented
Week 6	Integration and testing	Both	End-to-end workflow tested
Week 7	Report preparation	Both	Report drafted and reviewed

Appendix H – User/Industry Feedback

To be completed if formal user or industry feedback is collected.

Appendix I – Publications / Patents / Competitions / Product Outcomes

To be updated if the project is submitted to a competition, publication, patent process or product-development activity.

Appendix J – Individual Contribution Evidence

The project contribution split is 50% for Vasantha Nithi D Y, responsible for Intelligent ML Model Recommendation, and 50% for Saravana Yogesh M, responsible for Explainable AI & Model Insights. Evidence may include source-code commits, notebook sections, screenshots, experiment logs, test records and report sections attributable to each member.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome	Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	Problem definition, requirements and engineering gap	SO1	Ch. 1, Ch. 2
Engineering design under constraints	Requirements, alternatives, trade-offs and system design	SO2	Ch. 3
Written/oral technical communication	Complete technical report and final presentation	SO3	Whole report + Viva
Ethics and professional responsibility	Ethics, responsible AI, data privacy and limitations	SO4	Ch. 3 (3.7)
Teamwork and leadership	Individual roles, milestone planning and contribution statement	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	Test plan, metrics, comparison and interpretation	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	New knowledge and individual reflections	SO7	Ch. 5 (5.3)
Standards	Applicable standards and responsible practices	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	Computational efficiency, responsible use and user impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	Risk register and mitigation strategies	SO2/SO4	Ch. 3 (3.7)
Security considerations	Dataset privacy and responsible data handling	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	End-to-end architecture and module interfaces	SO1/SO2	Ch. 3 (3.5)
Project planning and management	Milestones, roles and budget	SO5	Ch. 4 (4.8)
Individual contribution	Contribution statement and reflection	SO1–SO7	Contribution Statement + Ch. 4/5

Appendix K – Model Recommendation Decision Logic

The recommendation module can be implemented as a transparent scoring or ranking mechanism. First, the system collects the selected evaluation metrics for every candidate model. Second, it checks whether the dataset requires emphasis on a particular metric, such as recall for an important minority class. Third, it ranks the candidates using the declared policy. Finally, it reports the winning candidate together with the metrics of the nearest alternatives. This sequence makes the recommendation auditable.

A simple accuracy-first policy may be appropriate for balanced educational datasets, but it should not be hard-coded as a universal rule. A more robust design can calculate a composite score from normalized accuracy, precision, recall and F1, with weights selected according to the application. The chosen weights must be documented because changing them can change the recommended model.

Tie handling is also an engineering consideration. If two models have similar performance, the system can prefer the simpler or more interpretable model, provided the performance difference is within a declared tolerance. This avoids presenting insignificant metric differences as meaningful superiority.

Appendix L – Explainability Design Notes

Explainability outputs must be interpreted carefully. Feature importance indicates how much a model relies on a feature under the selected explanation method; it does not establish that changing that feature will change the real-world outcome. Correlated features can also distribute importance across variables, making individual rankings less definitive.

Global explanations are useful for understanding overall model behaviour, whereas local explanations can help inspect a particular prediction. A mature version of ML Insight Studio can provide both views and allow the user to compare them. The interface should also display the model version, dataset version and preprocessing configuration so that an explanation can be traced to the experiment that produced it.

For reproducibility, explanation generation should use a defined random state when the selected technique contains stochastic components. Explanation parameters should be stored with the experiment record. If explanations are unstable across runs, this instability should be reported rather than hidden.

Appendix M – Dataset Quality Checklist

Before model training, the user should verify that the target variable exists, that duplicate records are handled appropriately, that missing values are identified, and that feature types are correctly detected. The dataset should also be inspected for unexpected constant columns, extreme values and incompatible text fields.

The target distribution should be examined because severe class imbalance can make accuracy misleading. When imbalance is detected, the evaluation should include class-sensitive metrics. Resampling or class weighting may be considered, but the choice must be documented because it changes the learning problem.

Potential data leakage should be checked carefully. Information that would not be available at prediction time should not be used as a training feature. Preprocessing transformations should be fitted only on training data when the evaluation design requires strict separation.

Appendix N – Testing Scenarios

Normal input test: provide a valid structured classification dataset and verify that all modules execute in sequence. The expected result is a complete model comparison, recommendation and explanation output.

Missing-value test: introduce missing values in selected numerical and categorical columns and verify that the preprocessing stage handles them according to the configured strategy.

Categorical-feature test: provide text-based categories and verify that encoding produces a compatible numerical representation without inconsistent data types.

Small-dataset test: provide a small dataset and verify that the system either completes training or reports a clear limitation instead of producing misleading results.

Class-imbalance test: provide an imbalanced target and verify that precision, recall and F1 are displayed so that accuracy is not interpreted in isolation.

Repeatability test: execute the same configuration multiple times and verify that the defined random-state procedure produces consistent outputs.

Appendix O – Suggested Final Demonstration Sequence

The final demonstration should begin by introducing the engineering problem: users need help selecting an appropriate machine learning model and understanding its predictions. The dataset is then loaded into the platform, followed by a short inspection of its dimensions, feature types and target distribution.

The next stage demonstrates preprocessing and candidate model training. The model comparison table should be shown before the recommendation so that the evaluator can see the evidence used by the selection mechanism. The recommended model should then be highlighted with its principal evaluation metrics.

The demonstration concludes with the Explainable AI module. The feature-importance or model-insight visualization should be shown and interpreted carefully. Finally, the limitations of automated recommendation and explainability should be stated to demonstrate responsible engineering judgment.